

Packet Tracer - Configure a Wireless Network

Objectives

- = Connect to a wireless router
- = Configure the wireless router
- = Connect a wired device to the wireless router
- = Connect a wireless device to the wireless router
- = Add an AP to the network to extend wireless coverage
- = Update default router settings

Introduction

In this activity, you will configure a wireless router and an access point to accept wireless clients and route IP packets. Furthermore, you will also update some of the default settings.

Instructions

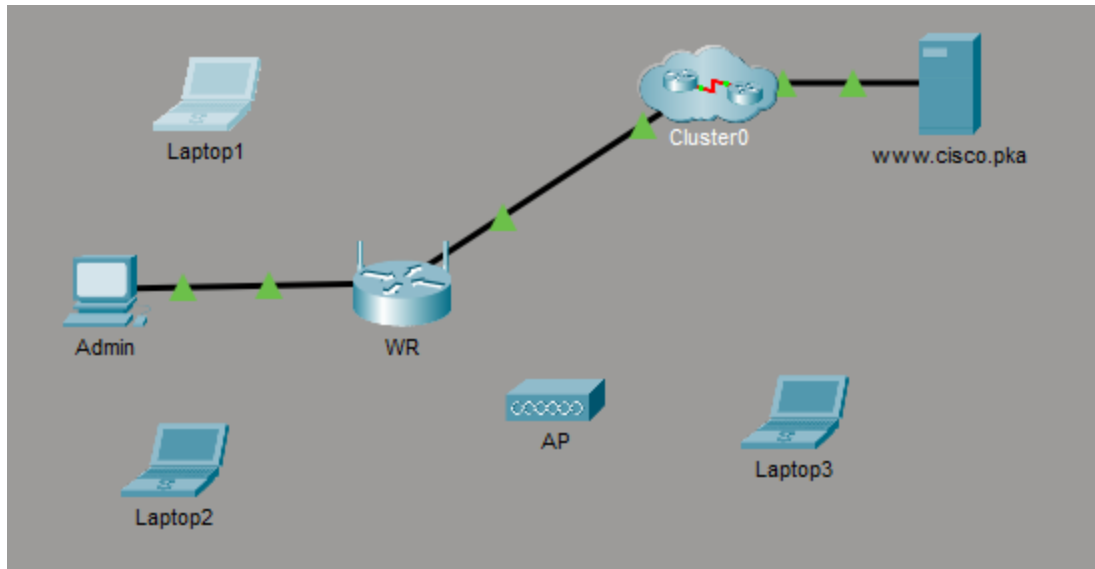
Part 1: Connect to a Wireless Router

Step 1: Connect Admin to WR.

- a. Connect **Admin** to **WR** using a straight-through Ethernet cable through the Ethernet ports. Select **Connections**, represented by a lightning bolt, from the bottom-left side of Packet Tracer. Click **Copper Straight-Through**, represented by a solid black line.
- b. When the cursor changes to connection mode, click **Admin** and choose **FastEthernet0**. Click **WR** and choose an available Ethernet port to connect the other end of the cable.

WR will act as a switch to the devices connected to the LAN and as a router to the internet. **Admin** is now connected to the LAN (**GigabitEthernet 1**). When Packet Tracer displays green triangles on both sides of the connection between **Admin** and **WR**, continue to the next step.

Note: If no green triangles are shown, make sure to enable **Show Link Lights** under **Options > Preferences**. You may also click **Fast Forward Time** just above the **Connections** selection box in the yellow bar.

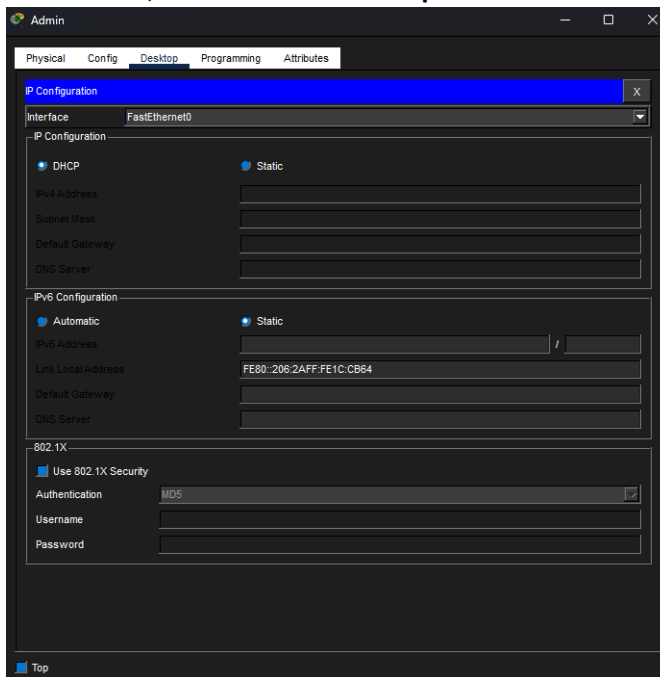


ทำการต่อสาย admin กับ WR

Step 2: Configure Admin to use DHCP.

To reach the **WR** management page, **Admin** must communicate on the network. A wireless router usually includes a DHCP server, and the DHCP server is usually enabled by default on the LAN. **Admin** will receive IP address information from the DHCP server on **WR**.

- Click **Admin**, and select the **Desktop** tab.



ตั้งค่าให้ Admin ใช้ DHCP

b. Click **IP Configuration** and select **DHCP**.

Questions:

What is the IP address of the computer?

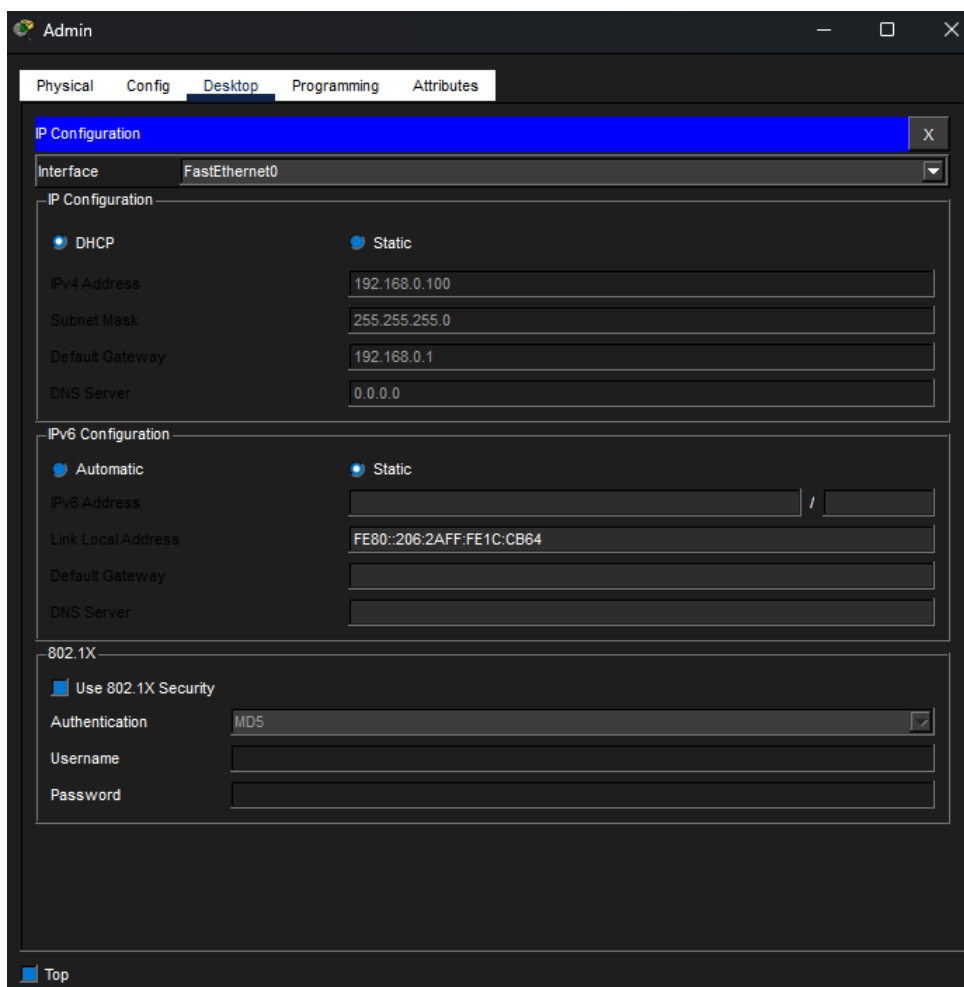
คำตอบ 192.168.0.100

What is the subnet mask of the computer?

คำตอบ 255.255.255.0

What is the default gateway of the computer?

คำตอบ 192.168.0.1

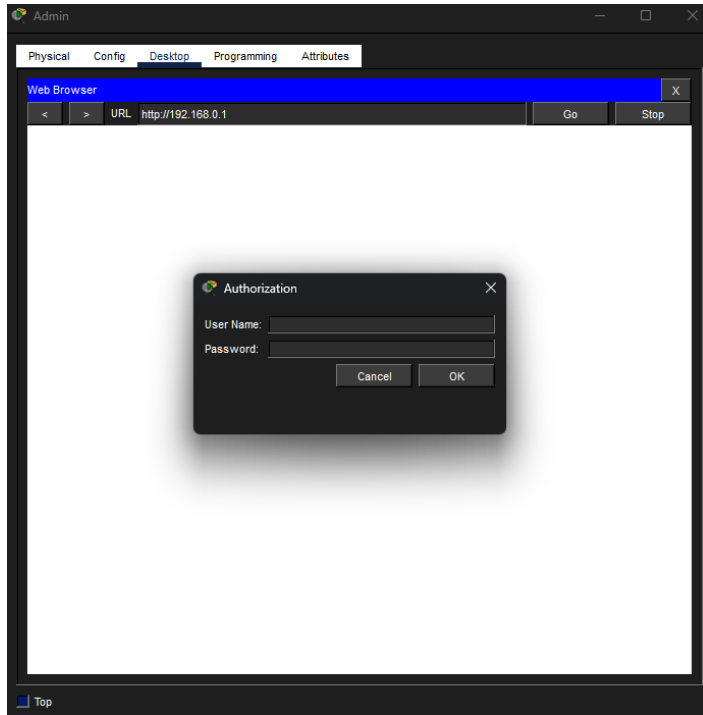


c. Close the **IP Configuration** window.

Note: Values can vary within the network range due to normal DHCP operation.

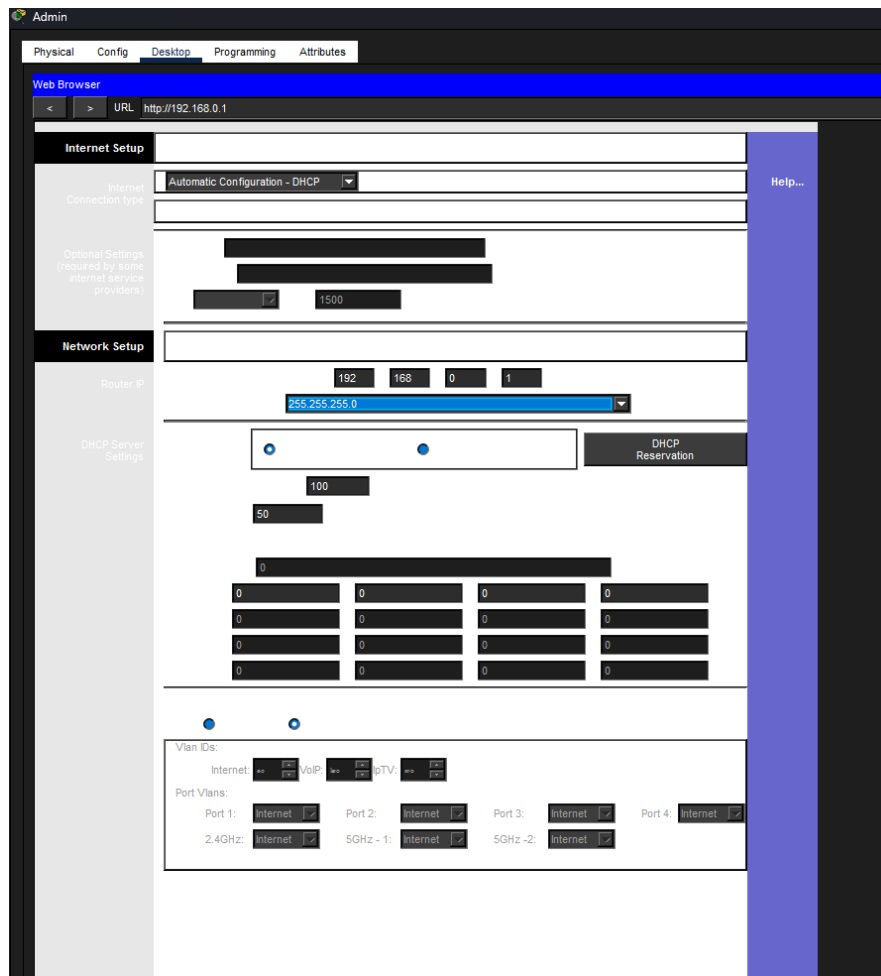
Step 3: Connect to the WR Web Interface.

- a. In the **Desktop** tab on **Admin**, choose **Web Browser**.
- b. Enter **192.168.0.1** in the URL field to open the web configuration page of the wireless router.



ทำการเปิด 192.168.0.1 ที่ Admin

- c. Use **admin** for both the username and password.
- d. Under the Network Setup heading on the **Basic Setup** page, notice the IP address range for the DHCP server.



Question:

Is the IP address for **Admin** within this range? Is it expected? Explain your answer.

คำตอบ IP ของเครื่อง Admin จะอยู่ในช่วง 192.168.0.1

Step 4: Configure the Internet Port of WR.

In this step, **WR** is configured to route the packets from the wireless clients to internet. You will configure the **Internet** port on **WR** to connect to the internet.

- Under the **Internet Setup** at the top of the **Basic Setup** page, change the Internet IP address method from **Automatic Configuration – DHCP** to **Static IP**.
- Type the IP address to be assigned to the Internet interface as follows:

Internet IP Address: 209.165.200.225

Subnet Mask: 255.255.255.252

Default Gateway: 209.165.200.226

DNS Server: 209.165.201.1

c. Scroll down the page and click **Save Settings**.

The screenshot shows the Cisco Admin interface with the 'Config' tab selected. The 'Internet Setup' section is active, showing the 'Static IP' connection type. The IP address fields are set to 209.165.200.225, the subnet mask to 255.255.255.252, and the default gateway to 209.165.200.226. The DNS server is set to 209.165.201.1. The 'Network Setup' section is also visible, showing the router IP as 192.168.0.1 and the DHCP server settings.

Admin

Physical Config Desktop Programming Attributes

Web Browser

URL: http://192.168.0.1/apply.cgi

Setup

Wireless To-Band Home Router

Internet Setup

Internet Connection type: Static IP

Optional Settings (required by some internet service providers)

Network Setup

Router IP: 192.168.0.1

DHCP Server Settings

Vlan IDs:

ทำการเพิ่มดังนี้

Internet IP Address: 209.165.200.225

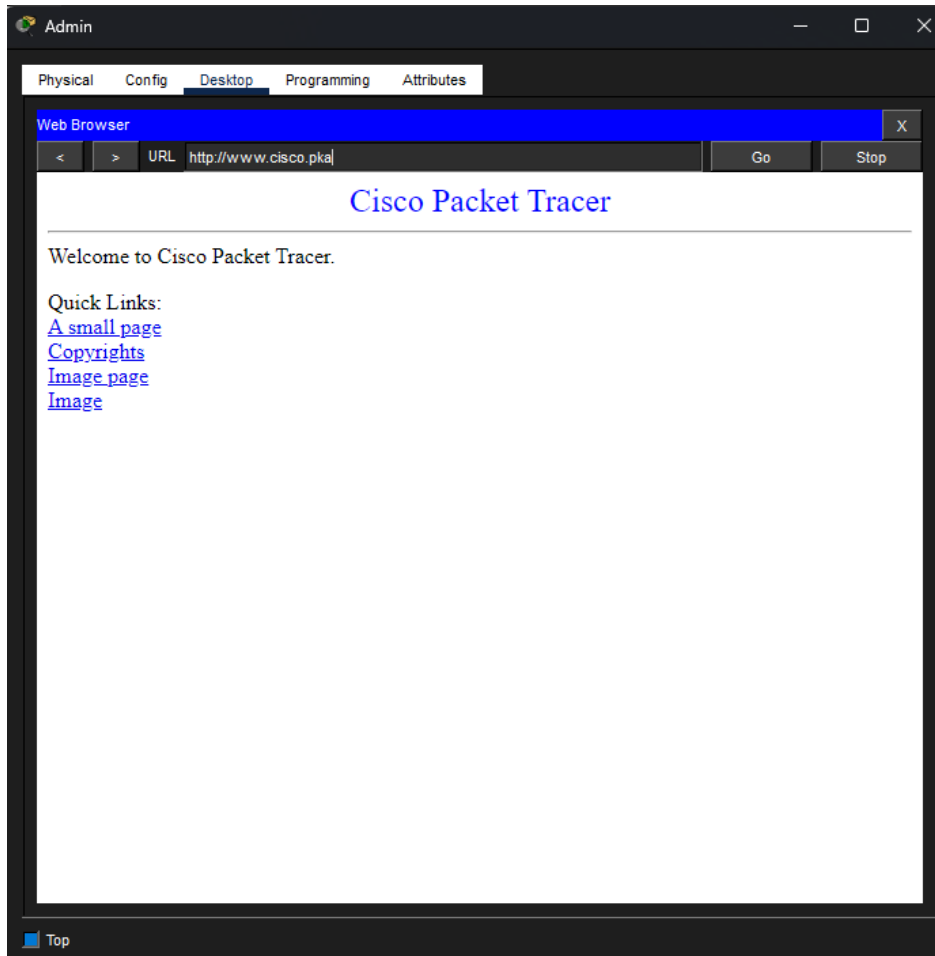
Subnet Mask: 255.255.255.252

Default Gateway: 209.165.200.226

DNS Server: 209.165.201.1

Note: If you get a **Request Timeout** message, close the Admin window and wait for the orange lights to turn into green triangles. Click the fast forward button to make this happen faster. Then reconnect to **WR** from **Admin's** browser using the process explained in Step 3.

d. To verify connectivity, open a new web browser and navigate to **www.cisco.pka** server.



สามารถเข้าเว็บได้แสดงว่าตั้งค่าถูกต้อง

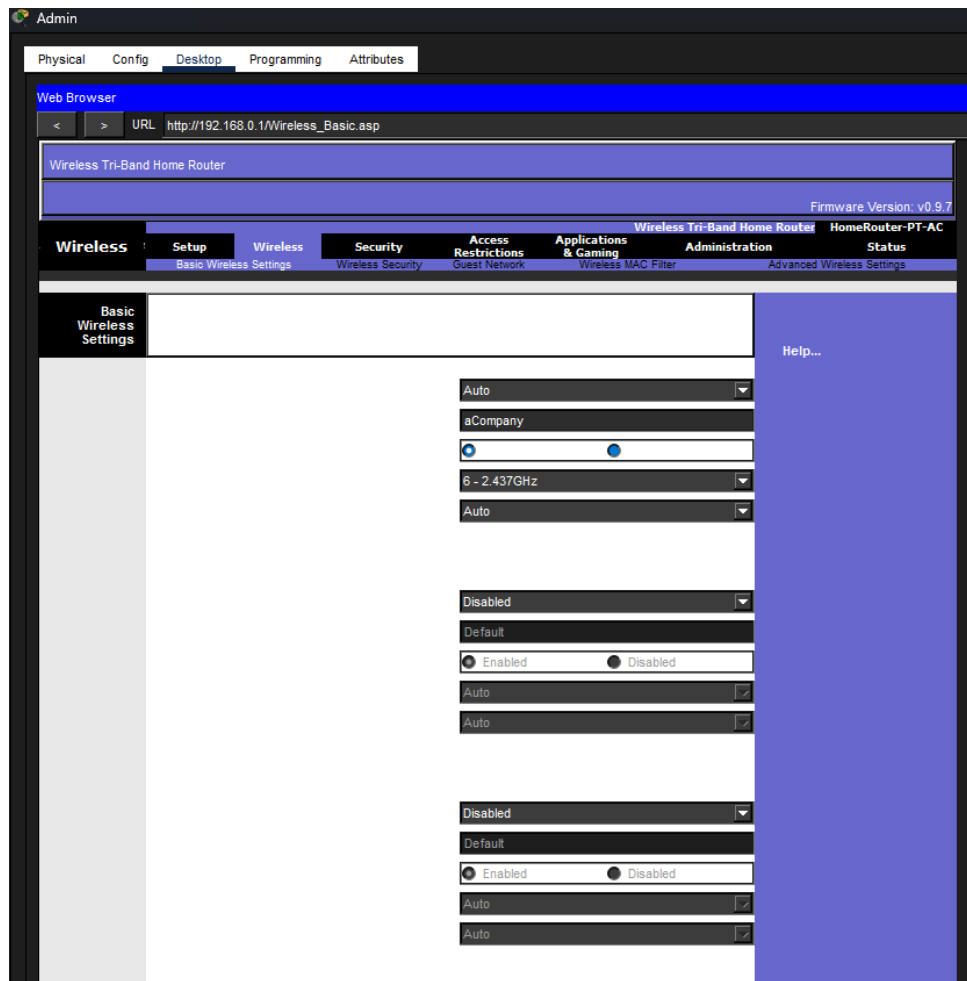
Note It may take a few seconds for the network to converge. Click **Fast Forward Time** or **Alt+D** to speed up the process.

Part 2: Configure the Wireless Settings

In this activity, you will only configure the wireless settings for 2.4 GHz.

Step 1: Configure the WR SSID.

- Navigate to the **WR** GUI interface at **192.168.0.1** in a web browser on **Admin**.
- Navigate to **Wireless > Basic Wireless Settings**.
- Change **Network Name (SSID)** to **aCompany** for only 2.4 GHz. Notice that SSIDs are case-sensitive.
- Change the **Standard Channel** to **6 - 2.437GHz**.
- For this activity, disable both 5 GHz frequencies. Leave the rest of the settings unchanged.
- Scroll to the bottom of the window and click **Save Settings**.

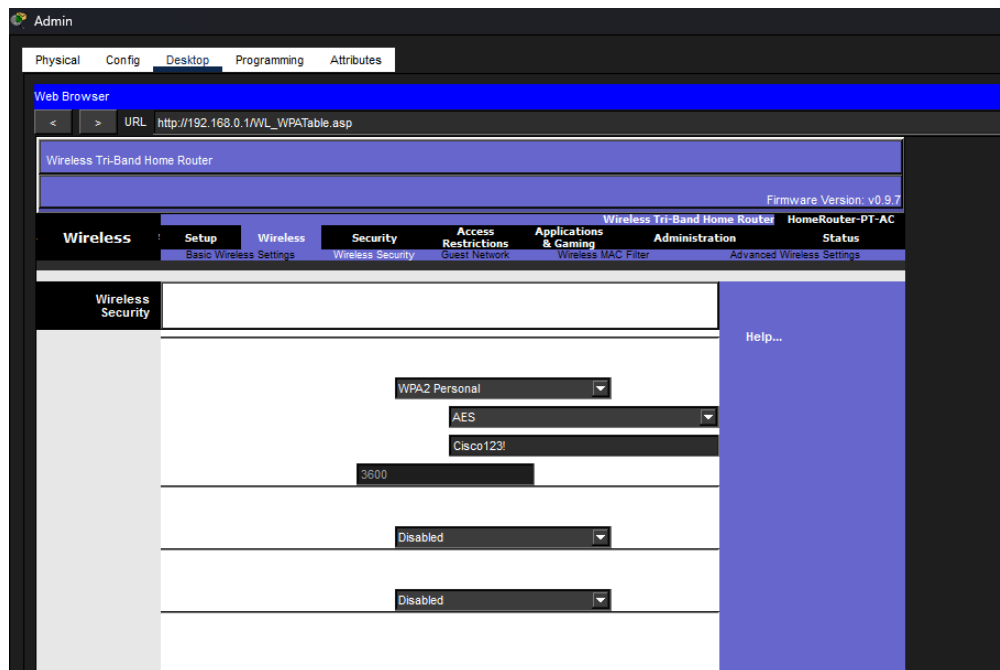


ทำการตั้งค่าชื่อ SSID คือ aCompany และตั้งค่า Standard Channel เป็น 6 - 2.437GHz.

Step 2: Configure wireless security settings.

In this step, you configure the wireless security settings using WPA2 security mode with encryption and passphrase.

- Navigate to **Wireless > Wireless Security**.
- Under the 2.4 GHz heading, select **WPA2 Personal** for the Security Mode.
- For the Encryption field, keep the default **AES** setting.
- In the Passphrase field, enter **Cisco123!** as the passphrase.
- Click **Save Settings**.



ตั้งค่ารหัสให้เป็น Cisco123!

f. Verify that the settings in the **Basic Wireless Settings** and **Wireless Security** pages are correct and saved.

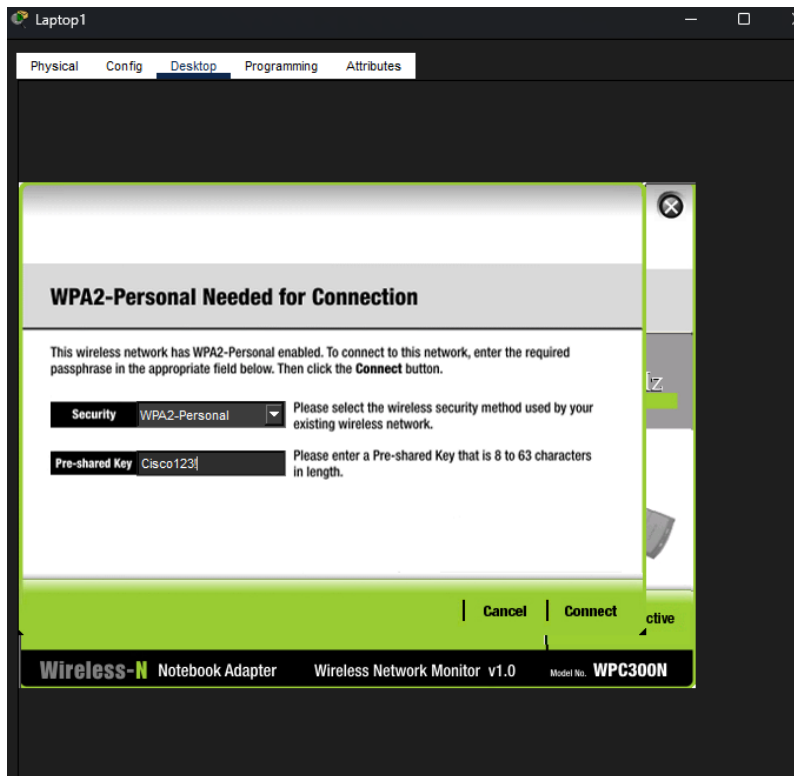
Step 3: Connect the Wireless Clients.

- a. Open **Laptop1**. Select **Desktop** tab. Click **PC Wireless**.
- b. Select the **Connect** tab. Click **Refresh** as necessary. Select the Wireless Network Name **aCompany**.

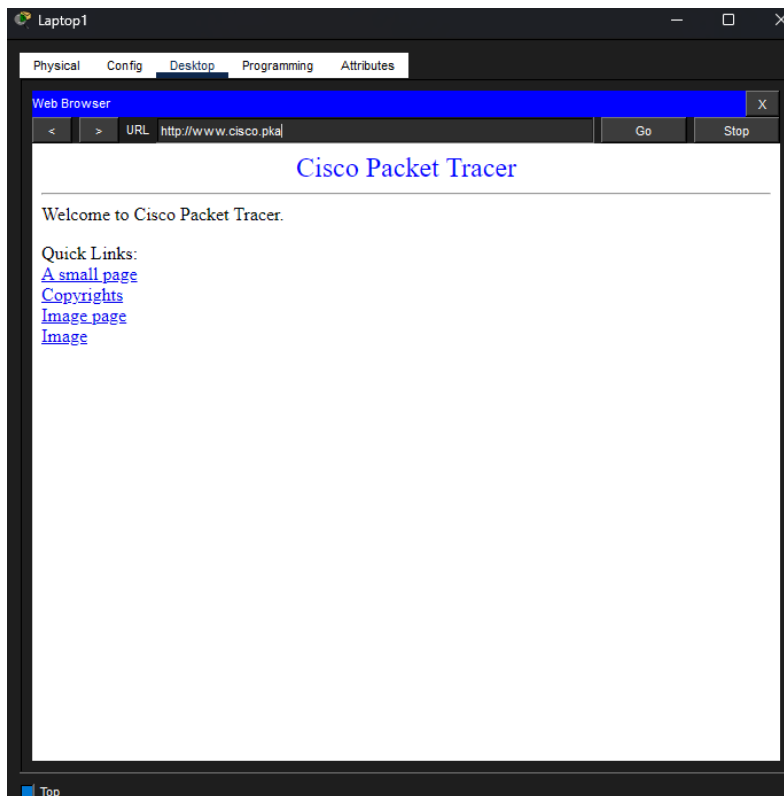


ทำการเชื่อมต่อ aCompany.

- c. Enter the passphrase configured in the previous step. Enter **Cisco123!** In the pre-shared key field and click **Connect**. Close the PC Wireless window.

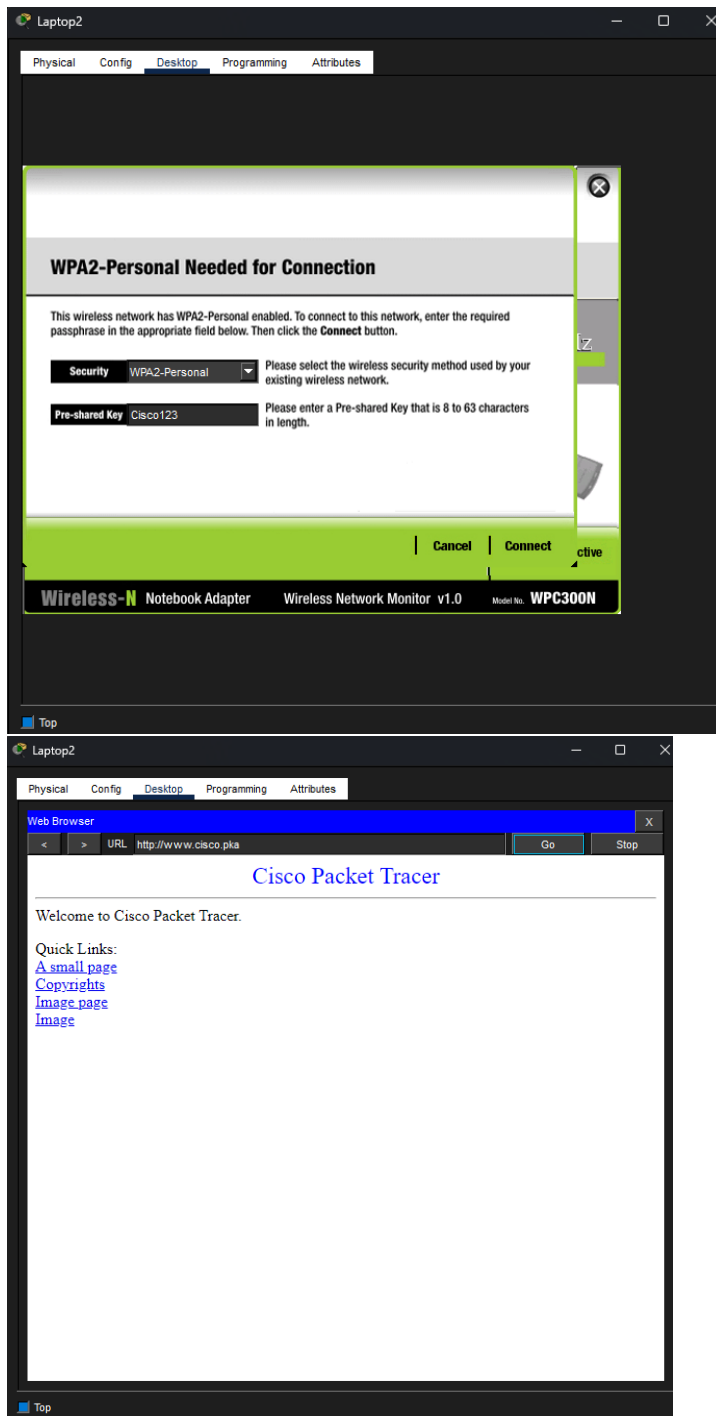


d. Open a web browser and verify that you can navigate to **www.cisco.pka** server.



สามารถเข้าเว็บได้แสดงว่าตั้งค่าถูกต้อง

e. Repeat the above steps to connect **Laptop2** to the wireless network.



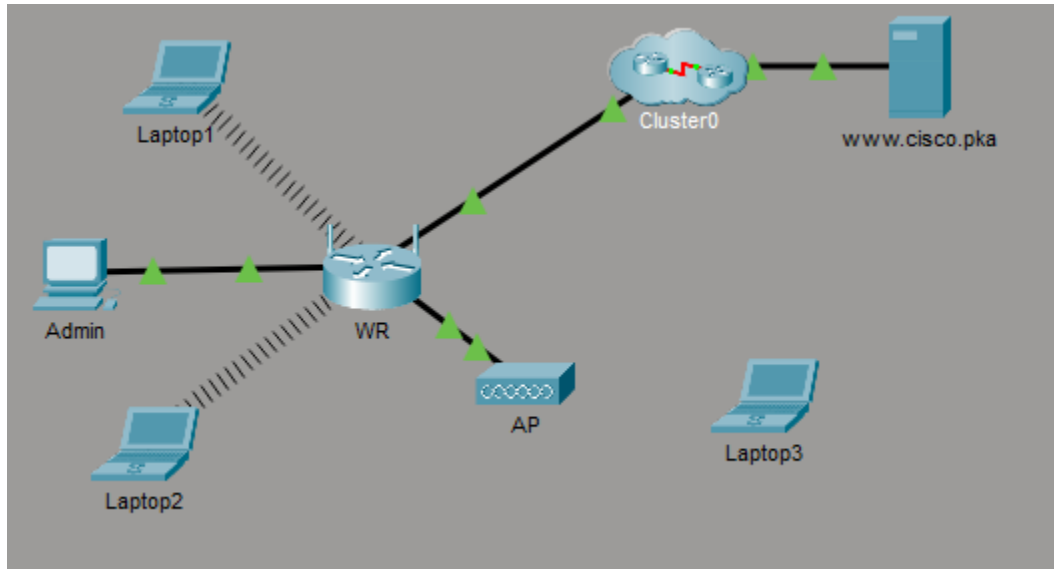
ตั้งค่าใน Laptop 2 และสามารถเข้าเว็บได้แสดงว่าตั้งค่าถูกต้อง

Part 3: Connect Wireless Clients to an Access Point

An access point (AP) is a device that extends the wireless local area network. An access point is connected to a wired router using an Ethernet cable to project the signal to a desired location.

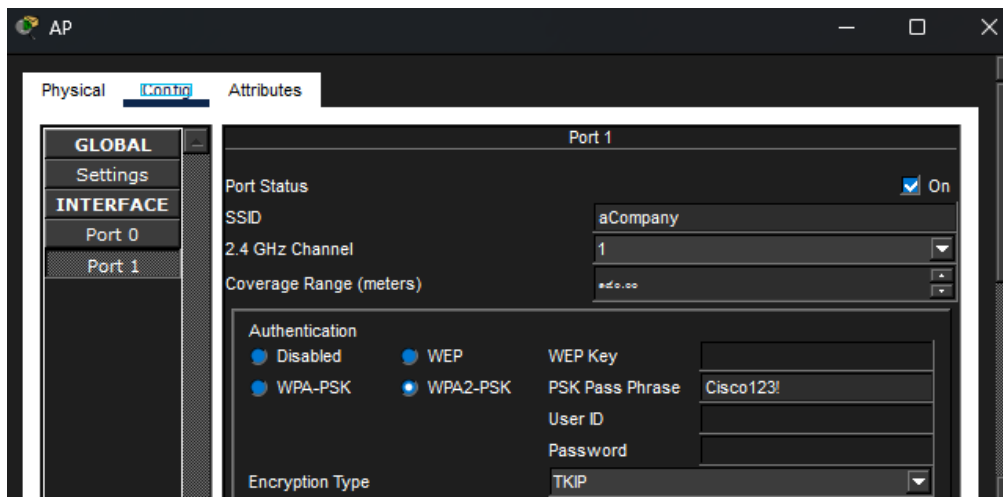
Step 1: Configure the Access Point.

- a. Connect **Port 0** of **AP** to an available Ethernet port of **WR** using a straight-through Ethernet cable.



ทำการเชื่อมต่อ AP กับ WR

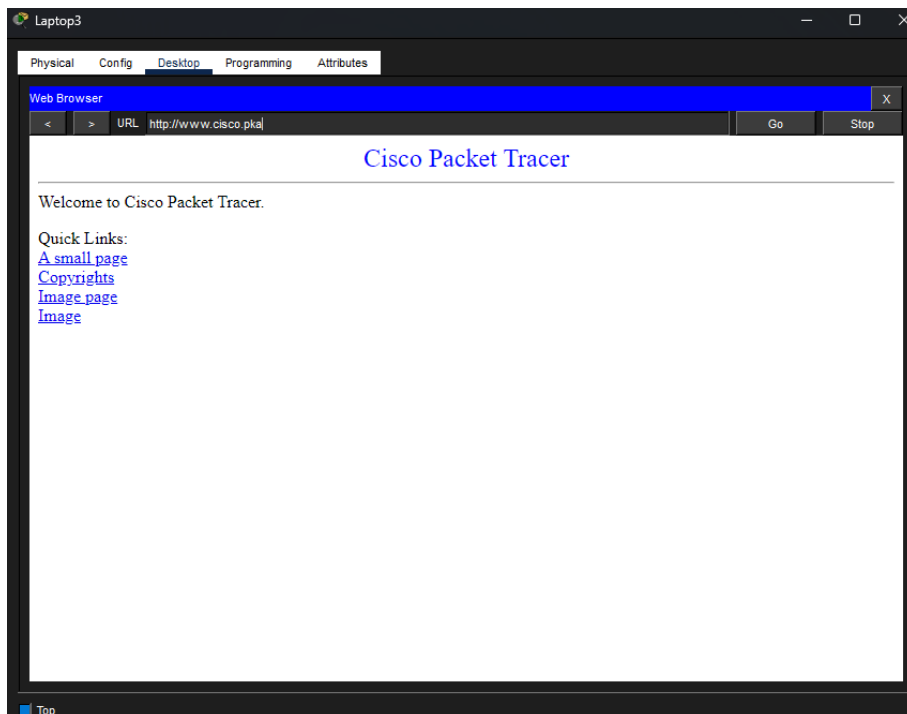
- b. Click **AP**. Select the **Config** tab.
- c. Under the INTERFACE heading, select **Port 1**.
- d. In the SSID field, enter **aCompany**.
- e. Select **WPA2-PSK**. Enter the passphrase **Cisco123!** in the Pass Phrase field.
- f. Keep **AES** as the default Encryption Type.



ทำการตั้งค่าเครื่อง AP

Step 2: Connect the Wireless Clients.

- Open **Laptop3**. Select **Desktop** tab. Click **PC Wireless**.
- Select the **Connect** tab. Click **Refresh** as necessary. Select the Wireless Network Name **aCompany** with the stronger signal (Channel 1) and click **Connect**.
- Open a web browser and verify that you can navigate to **www.cisco.pka** server.

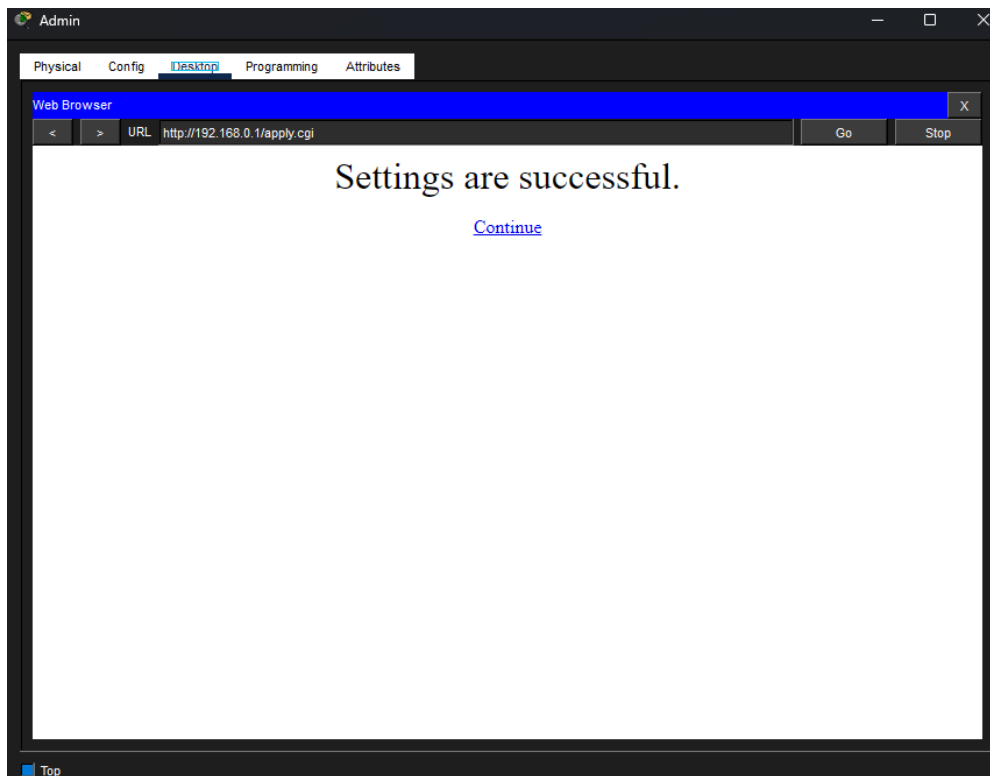


สามารถเข้าเว็บได้แสดงว่าตั้งค่าถูกต้อง

Part 4: Other Administrative Tasks

Step 1: Change the WR Access Password.

- a. On **Admin**, navigate to WR GUI interface at **192.168.0.1**.
- b. Navigate to **Administration > Management** and change the current **Router Password** to **cisco**.
- c. Scroll to the bottom of the window and click **Save Settings**.
- d. Use the username **admin** and the new password **cisco** when prompted to log in to the wireless router. Click **OK** to continue.
- e. Click **Continue** and move on to the next step.



Step 2: Change the DHCP address range in WR.

In this step, you will change the internal network address from 192.168.0.0/24 to 192.168.50.0/24. When the LAN network address changes, the IP addresses on the devices in the LAN and WLAN must be renewed to receive new IP addresses before the lease is timed out.

- a. Navigate to **Setup > Basic Setup**.
- b. Scroll down the page to **Network Setup**.
- c. The IP address assigned to **Router IP** is 192.168.0.1. Change it to 192.168.50.1. Verify that IP address still start at .100, and there are 50 available IP addresses in the DHCP pool.
- d. Add **209.165.201.1** as the DNS server with the DHCP settings.

- e. Scroll to the bottom of the window and click **Save Settings**.
- f. Note that the DHCP range of addresses has been automatically updated to reflect the interface IP address change. The Web Browser will display a **Request Timeout** after a short time.

Question:

Why?

เพราะว่าเพิ่งเปลี่ยน IP ของเราเตอร์ จาก 192.168.0.1 เป็น 192.168.50.1 ในขณะที่เครื่อง Admin ยังคงใช้ IP เดิมในวง 192.168.0.x ทำให้เครื่อง Admin และเราเตอร์อยู่คนละวงเครือข่ายกัน จึงไม่สามารถสื่อสารกันได้และเกิด Request Timeout

- g. Close the **Admin** web browser.
- h. In **Admin Desktop** tab, click **Command Prompt**.
- i. Type **ipconfig /renew** to force **Admin** re-acquire its IP information via DHCP.

```
C:\>ipconfig /renew

IP Address.....: 192.168.50.106
Subnet Mask.....: 255.255.255.0
Default Gateway...: 192.168.50.1
DNS Server.....: 209.165.201.1

C:\>
C:\>
```

Question:

What is the new IP address information for **Admin**?

ข้อมูลที่ได้รับใหม่จะเป็นวง 50 ตามที่ตั้งค่าไว้ได้แก่

- IP Address: 192.168.50.106
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.50.1
- DNS Server: 209.165.201.1

j. Verify that you can still navigate to **www.cisco.pka** server.

k. Renew the IP address on other laptops to verify that you can still navigate to **www.cisco.pka** server.

l. Notice that **Laptop1** connected to the **AP** instead of **WR**.

Question:

Why?

เนื่องจากทั้งเครื่อง **AP** และ **WR** มีการปล่อยสัญญาณ Wi-Fi ในชื่อ เดียวกันคือ **aCompany** และตั้งรหัสผ่านเหมือนกัน แต่อุปกรณ์ **Laptop1** อยู่ในตำแหน่งที่ได้รับสัญญาณจาก **AP** แรงกว่าสัญญาณที่มาจาก **WR** ดังนั้นอุปกรณ์ไร้สายจึงฉลาดพอที่จะเลือกโรมมิ่ง ไปเกาะกับจุดกระจายสัญญาณที่มีความแรง ดีที่สุดโดยอัตโนมัติ